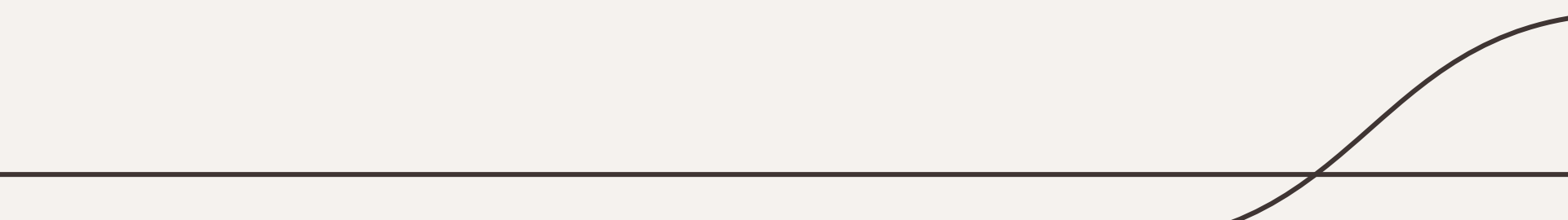


F1 Data

Robyn Luo And Aria Blondeau



Problem:

- There are many factors that go into producing a fast lap time in F1
 - Our focus is on tyre wear and analysing what effect the teams, tracks, and compounds have on the tyre wear
 - We looked at data across every race since 2022 to find these answers and more
-

Our Data:

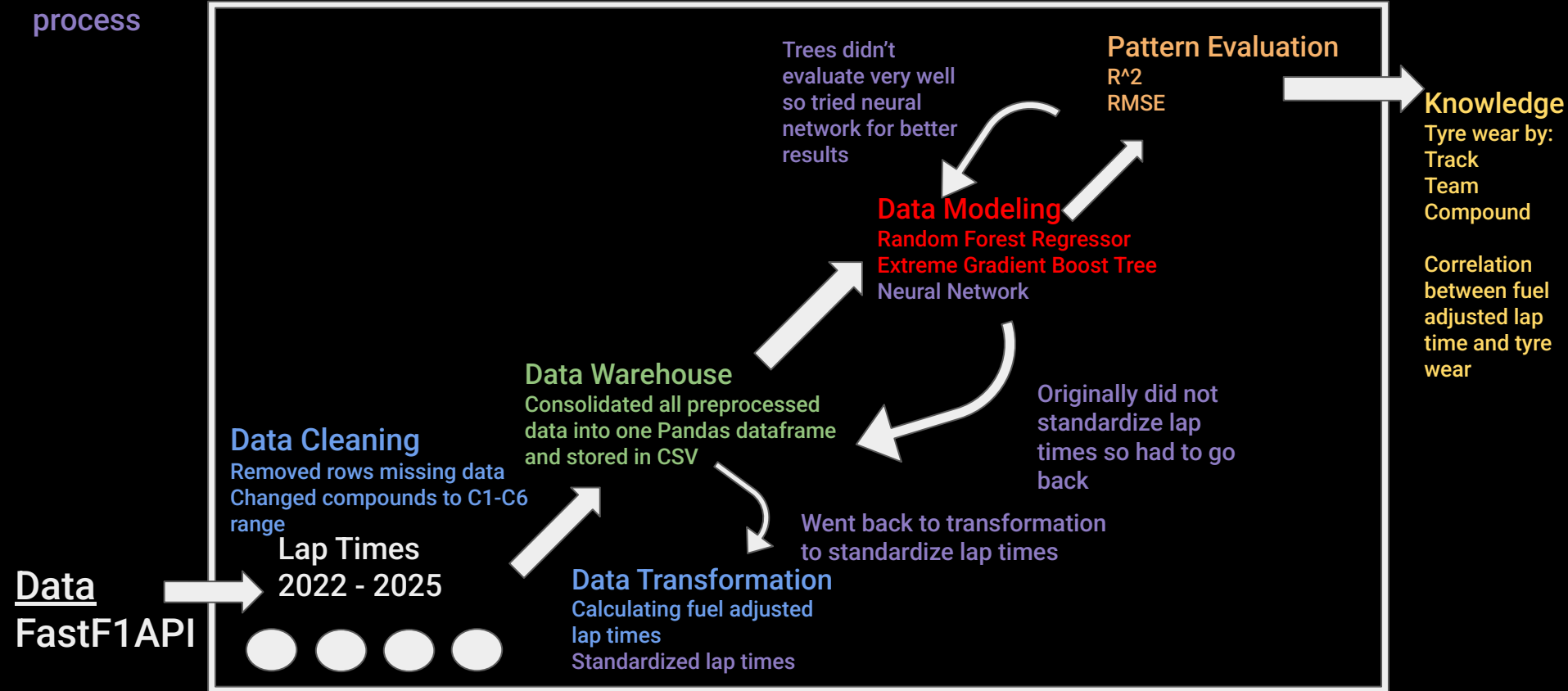
- We used the API: <https://docs.fastf1.dev/index.html>
- Size of our data is 300k rows of laps each with 22 features including:
 - Lap times
 - Tyre Compounds
 - Year/Session
 - Driver
 - And more



0	Time	Driver	DriverNumber	LapTime	LapNumber	Stint	PitOutTime	PitInTime	Sector1Time	...	IsAccurate	Year	Round	Session	fuel_kg	fuel_adjustment	fuel_effect	lap_time_s	fuel_delta_to_55	lap_time_fuel_std_55kg
0	0 days 01:04:15.340000	VER	1	0 days 00:01:40.236000	1.0	1.0	NaN	NaN	NaN	...	False	2022	1	Race	110.000000	3.300000	0.03	100.236	1.650000	98.586000
1	0 days 01:05:53.220000	VER	1	0 days 00:01:37.880000	2.0	1.0	NaN	NaN	0 days 00:00:31.285000	...	True	2022	1	Race	108.049107	3.241473	0.03	97.880	1.591473	96.288527
2	0 days 01:07:31.577000	VER	1	0 days 00:01:38.357000	3.0	1.0	NaN	NaN	0 days 00:00:31.499000	...	True	2022	1	Race	106.098214	3.182946	0.03	98.357	1.532946	96.824054
3	0 days 01:09:10.143000	VER	1	0 days 00:01:38.566000	4.0	1.0	NaN	NaN	0 days 00:00:31.342000	...	True	2022	1	Race	104.147321	3.124420	0.03	98.566	1.474420	97.091580
4	0 days 01:10:49.020000	VER	1	0 days 00:01:38.877000	5.0	1.0	NaN	NaN	0 days 00:00:31.498000	...	True	2022	1	Race	102.196429	3.065893	0.03	98.877	1.415893	97.461107
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
16	0 days 02:13:00.598000	PIA	81	0 days 00:01:34.588000	46.0	2.0	NaN	NaN	0 days 00:00:26.951000	...	True	2025	22	Race	9.668367	0.193367	0.02	94.588	-0.906633	95.494633
17	0 days 02:14:34.684000	PIA	81	0 days 00:01:34.086000	47.0	2.0	NaN	NaN	0 days 00:00:26.763000	...	True	2025	22	Race	7.438776	0.148776	0.02	94.086	-0.951224	95.037224
18	0 days 02:16:08.849000	PIA	81	0 days 00:01:34.165000	48.0	2.0	NaN	NaN	0 days 00:00:26.769000	...	True	2025	22	Race	5.209184	0.104184	0.02	94.165	-0.995816	95.160816
19	0 days 02:17:43.315000	PIA	81	0 days 00:01:34.466000	49.0	2.0	NaN	NaN	0 days 00:00:26.922000	...	True	2025	22	Race	2.979592	0.059592	0.02	94.466	-1.040408	95.506408

Purple is changes made based on the cyclical process

Data Mining Pipeline



Data Cleaning

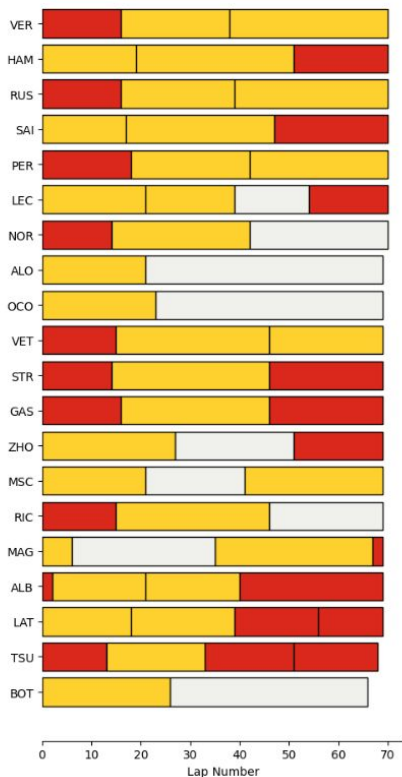
- Only used race data from 2022-2025 since new regulations were set
 - Practice/qualifying sessions have different run plans that wouldn't allow for easy degradation analysis
- Cleaned missing data by dropping rows (no suitable replacements)
- Removed laps that weren't driven during an normal track status (safety car, virtual safety car, red flag)
- Updated Tyre compounds to standard C1-C6 range instead of Soft, Medium, Hard
 - Allows us to compare the same tyre compounds across multiple tracks
- Isolated lap_time speed (for models):
 - Only accepted times $\leq 1.075 \times \text{median lap time}$
 - Removed data with rainy weather
- Adjusted times for fuel weight

Calculating And Standardizing Fuel Adjusted Times

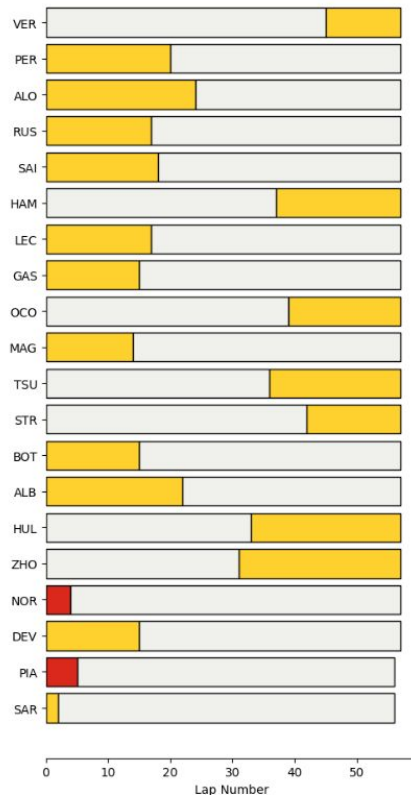
- As a Formula 1 car drives around the track, there are numerous components contributing to the lap time
- The primary elements that contribute to lap time is car weight and tyre degradation (there are many other factors but these are the major ones)
- Formula 1 cars have a max fuel load of 110kg at the beginning of the race and need at least 0.75kg of fuel remaining at the end for sampling, so assuming all cars start at 110kg we can then calculate the fuel consumption per lap and use an estimated weight effect to calculate the approximate change in lap time to a specified weight
 - The weight/fuel load data is not publicly available hence why we have to estimate, but based on numerous comparisons to other similar analyses these estimations are good enough
- We classify tracks into low, medium, and high speed categories as lower speed tracks normally carry a heavier penalty for more weight in the car and use this estimated fuel effect to calculate the change in lap time for the current weight of fuel in the car standardized to 55kg (the approximate average fuel load)
- Once we have the fuel adjusted lap time standardized to a specific weight, we can then look at the median lap time for a given weekend and calculate the percentage difference for each lap giving us a percent of tyre degradation in terms of lap time, which can then be used to compare degradation across different tracks

Looking At Tyre Strategies

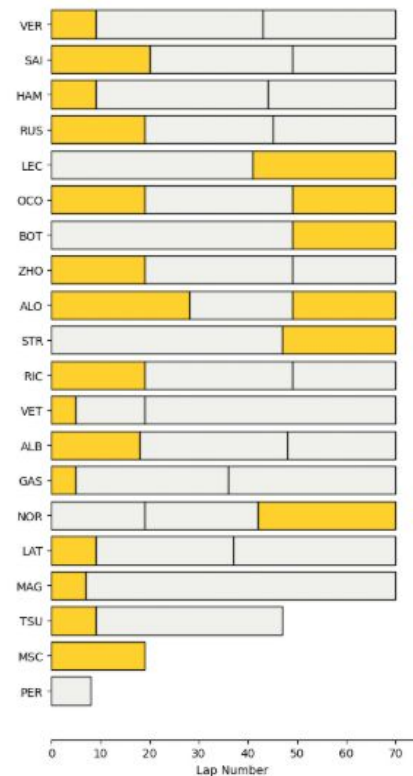
2022 Hungarian Grand Prix Strategies (Placement From Top to Bottom)



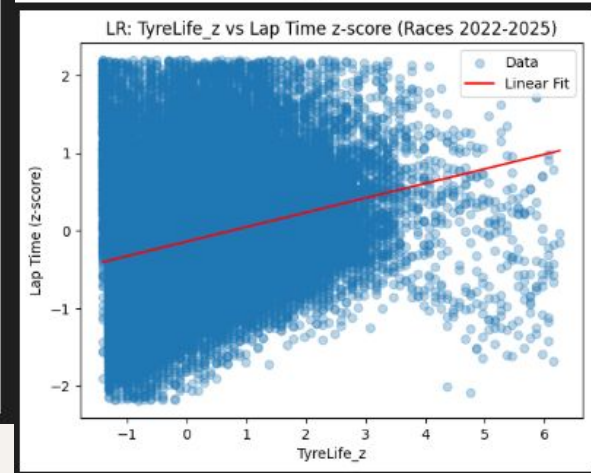
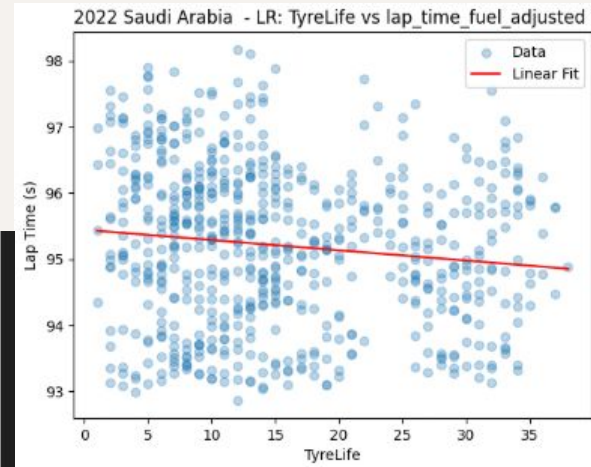
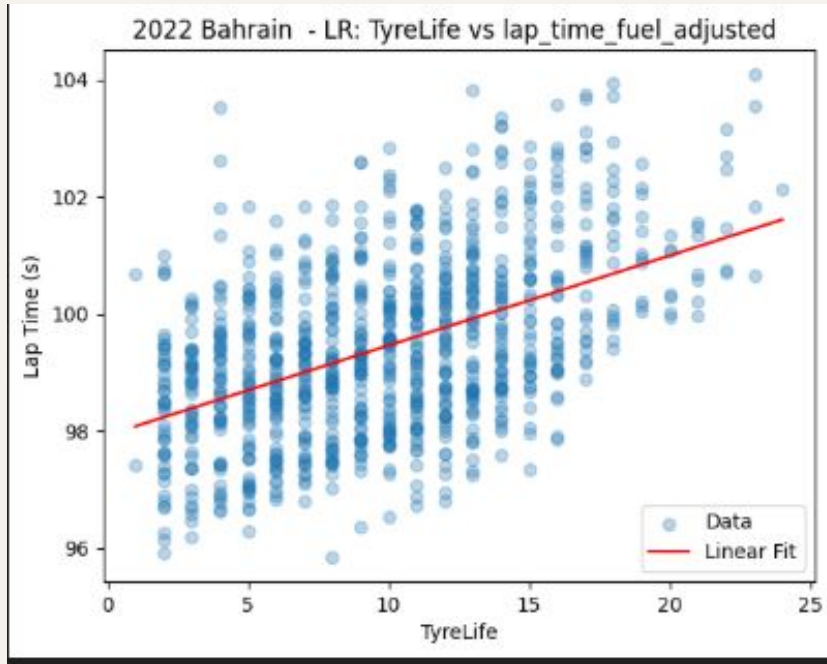
2023 Miami Grand Prix Strategies (Placement From Top to Bottom)



2022 Canadian Grand Prix Strategies (Placement From Top to Bottom)

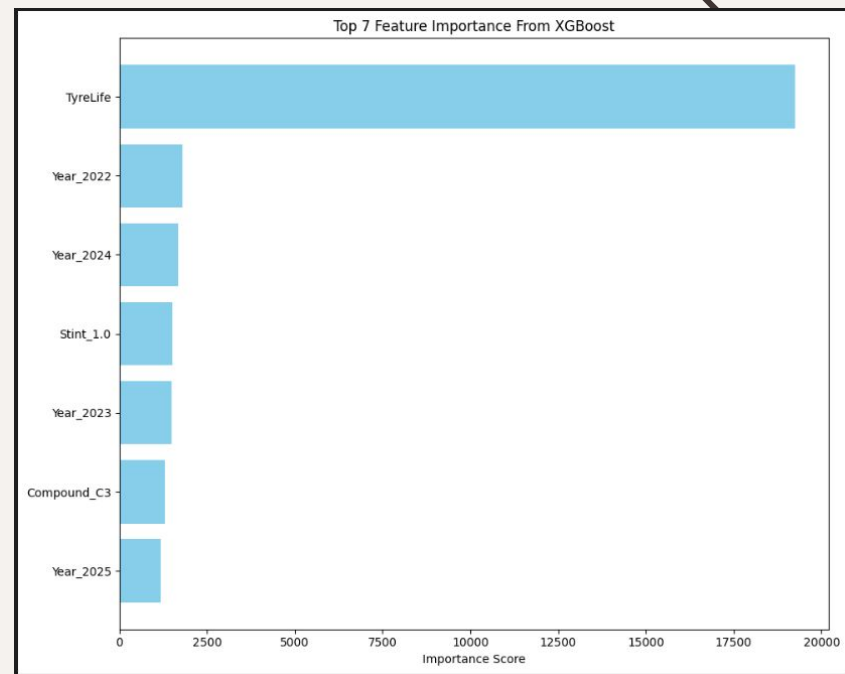


Tyre life vs Lap Time



Model

- Predicting Lap Time based on:
 - Numerical
 - TyreLife,
 - Categorical
 - Year, Round, Driver, Team,
 - Session, Tyre Compound, Position, Stint
- Random Forest Regressor
 - Many Independent trees
 - Good baseline
- XGBoost Tree (Extreme Gradient Boost)
 - Trains decision trees sequentially where each that improve upon the previous tree's errors
- Neural Network
 - Best Performance
 - Is able to find more patterns, but harder to interpret

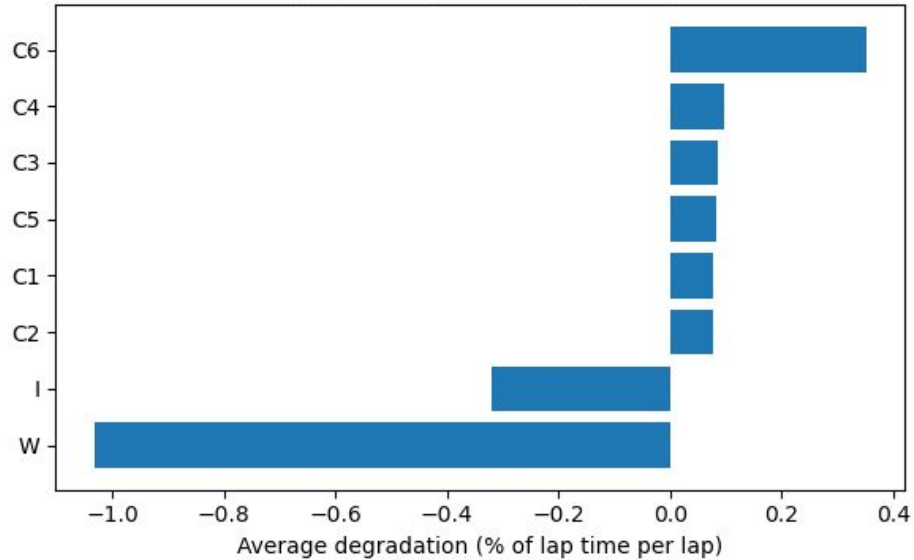


R^2 : 0.8718965588795043

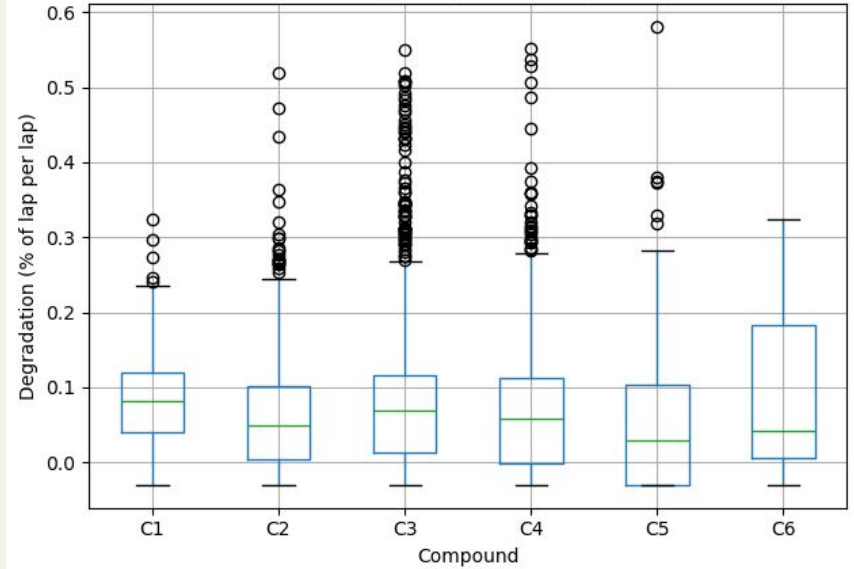
RMSE: 0.054633703637321215

Analysis

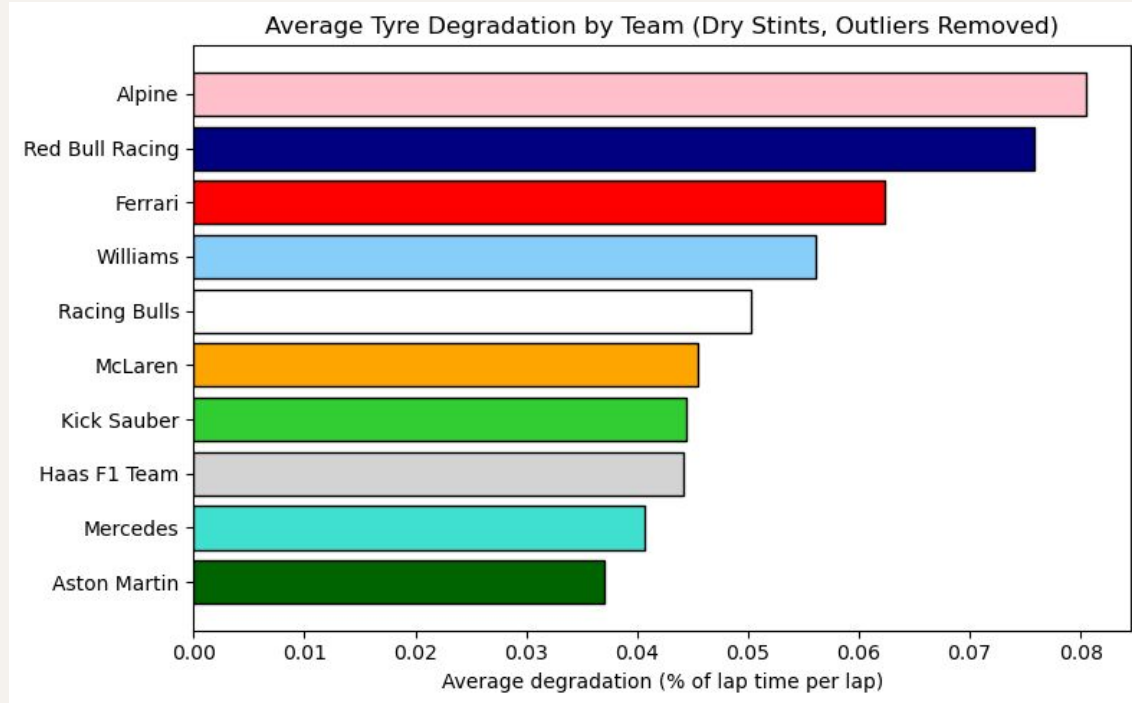
Average Tyre Degradation by Compound



Distribution of Tyre Degradation by Compound

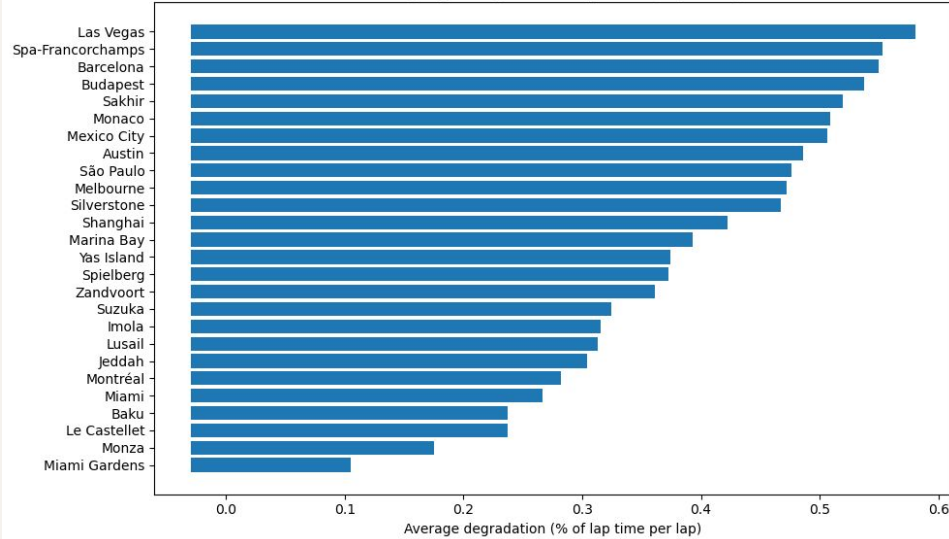


Analysis

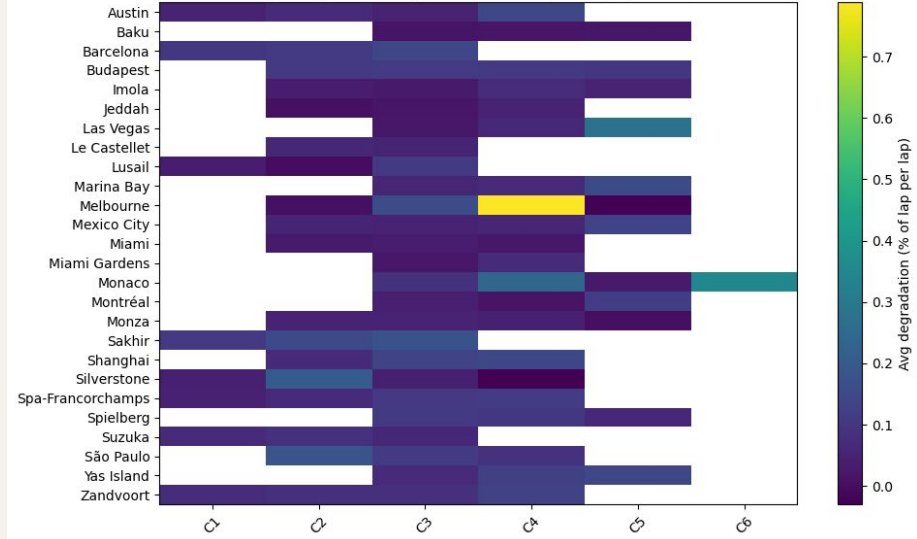


Analysis

Average Tyre Degradation by Track (Dry Stints)



Tyre Degradation by Track and Compound (Dry Stints)



Conclusion

- The data we found can be used by teams to influence strategies for future races as they can specifically look at tracks where they may deal with higher degradation and plan accordingly
 - Teams could also look at what teams perform better at certain tracks in terms of tyre degradation to try and glean insights for performance gains
- This data can also be used by Pirelli (the tyre manufacturer) or the FIA to influence decision-making around what tyres are brought to a race weekend for the soft, medium, and hard
 - If Pirelli sees a track that has unusually high degradation compared to what's normally expected they may need to bring more robust tyres to deal with the conditions

Challenges & Future Work

- One of the bigger struggles for this project was figuring out how to calculate the lap times adjusted for fuel
 - Since fuel and weight data is not publicly available, we ended up using estimations to calculate this
 - Future improvements could be made if we either had the weight data or were able to do a calculation to better estimate the fuel effect
- Another struggle is that even though we have data going back to 2018, since there was a regulation change between 2021 and 2022 the data is not really comparable between the different rules eras which limited the scope of our useful data
 - It would also be very interesting to be able to look into older data pre-2018 to compare multiple different regulation eras but unfortunately this would require more detailed data that will likely never exist for those years
- Future work that would be interesting to pursue could include looking to see how weather/temperature also affects tyre degradation, examining the differences between the types of degradation (wear, thermal, graining, and blistering) and how they affect lap time, and also breaking tracks down sector by sector to find the sections and turns that are the most punishing on tyres

The image features a light gray background with two horizontal dark gray lines, one near the top and one near the bottom. A dark gray curved line starts from the top right corner and extends towards the center of the slide.

Thank you!